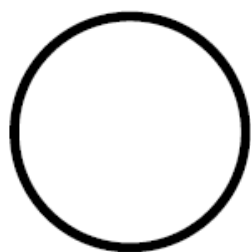
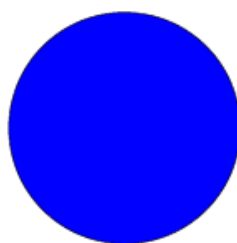


PROBLEMS-Angular Momentum

You are in lab and see in front of you a hollow cylinder spinning with $\omega = 20$ rpm. You decide to drop the solid cylinder on top of it.



Hollow Cylinder
 $m = 1.5$ kg



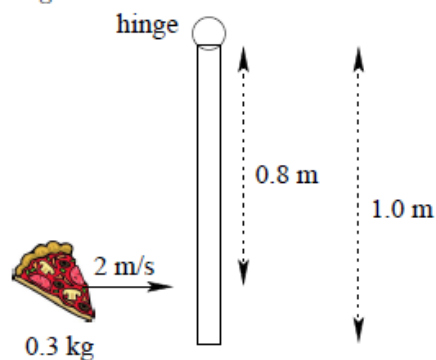
Solid Cylinder
 $M = 2$ kg

- (a) What is the final ω after you drop the solid disk?

PROBLEMS-Angular Momentum

You awake in the middle of the night to find that your slovenly roommate has ONCE AGAIN left the dorm room door open. Being too tired to get up and close the door, you grab a nearby piece of pizza and fling it at the door. It strikes the door as shown in the figure below and sticks to the door.

Treat the pizza as a point object that strikes the 1.5 kg door 0.8 meters away from the hinge.



- (a) Determine the angular velocity ω (in $\frac{rad}{sec}$) of the pizza-stained door just after the collision.
- (b) Determine the time for the pizza-stained door to rotate by $1/4$ turn. Assume no friction, so the pizza-stained door moves with constant ω .